

Review article

Functional requirements of automotive head-up displays: A systematic review of literature from 1994 to present



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ABSTRACT

Despite the long history of automotive head-up displays (HUDs), what information they should display in different situations to best serve the driver remains unanswered. The lack of understanding hinders designing useful automotive HUD systems. In an effort to address this, the current study investigated the developer, researcher and user perspectives on the functional requirements of automotive HUDs through literature review. The review results indicated that: 1) the existing commercial HUDs perform largely the same functions as the conventional in-vehicle displays, 2) past research studies proposed various HUD functions for improving driver situation awareness and driving safety, 3) autonomous driving and other new technologies are giving rise to new HUD information, and 4) little research is currently available on HUD users' perceived information needs. Based on the review results, this study provides insights into the functional requirements of automotive HUDs and also suggests some future research directions for automotive HUD design.

1. Introduction

Compared to traditional head-down displays (HDDs), automotive head-up displays (HUDs) reduce the driver's eye-off-the-road time (EoRT) (Gish et al., 1999; Horrey and Wickens, 2004; Liu et al., 2004; Medenica et al., 2011; Nwakacha et al., 2013; Palinko et al., 2013; Steinfeld and Green, 1995; Weinberg et al., 2011) and reaccommodation demands (Gish and Staplin, 1995) by presenting visual information within the driver's forward field of view (FoV), at a focal plane further into the forward scene. With the advantages in information access costs, automotive HUDs are considered to have the potential to improve driving performance and safety. Some studies have empirically demonstrated the positive effects of HUDs over HDDs in terms of performance of primary and secondary driving tasks (Gish et al., 1999; Liu and Wen, 2004; Srinivasan et al., 1994; Steinfeld and Green, 1995; Wittmann et al., 2006), and, driver distraction and workload (Weinberg et al., 2011).

Since the first introduction by General Motors in 1988, commercial automotive HUDs have been available in various production cars. The use of automotive HUDs is expected to increase in the years to come (Future Market Insights, 2015; IHS, 2013; MarketsandMarkets.com, 2016; MarketsandMarkets.com, 2015; Pala, 2012; Zion Market Research, 2016). It is projected that by 2024, almost one-third of all cars will be equipped with a HUD system (ABI Research, 2015).

The projection for increased use of automotive HUDs seems to be partly based on the fact that the range of possible automotive HUD functions is expanding with the advent of new technologies in the areas of photonics, augmented reality, internet-of-things and autonomous systems (Gabbard et al., 2014). Indeed, past research studies have proposed a variety of automotive HUD functions reflecting the technological advances. Some examples include displaying hazard warnings (Charissis and Papanastasiou, 2008; George et al., 2012; Lee et al., 2015; Maag et al., 2015; Park and Kim, 2013; Plavšić et al., 2009; Suzuki and Hashimoto, 2012; Tonnies and Klinker, 2006; Tonnies et al., 2005), traffic sign/signal notifications (Caird et al., 2008; Chang et al., 2015; Lee et al., 2015; Park and Kim, 2013; Yang et al., 2016) and driving instructions (Charissis and Papanastasiou, 2008; Lee et al., 2015; Lin et al., 2011; Maag et al., 2015; Park and Kim, 2013; Riener and Jeon, 2012; Riener and Thaller, 2014; Yoon et al., 2014). These proposed functions reflect the commonly shared idea that the usefulness of automotive HUD lies in enhancing driving performance and safety by supporting the primary and secondary driving tasks. Other studies, on the other hand, have proposed automotive HUD functions pertaining to non-driving-related tasks, such as displaying communication-related information (Charissis et al., 2007; Maurer et al., 2014; Wang et al., 2014; Zimmermann et al., 2014) and outside environment information (Fujimura et al., 2013), and, supporting augmented reality games (Schroeter et al., 2014; Steinberger et al., 2015). These proposals

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represent the more recent idea that automotive HUDs could be utilized to create new driver experience - they tend to involve displaying new types of information previously non-existent or difficult to display through HUDs.

The current technological feasibility of creating various HUD functions, however, does not mean that an automotive HUD system can be designed to display a wide variety of information without limit. Displaying too much information through HUDs can result in information overload. Also, poor interface design can cause problems, such as visual clutter (Gish and Staplin, 1995; Pausie, 2015), misaccommodation (Gish and Staplin, 1995; Ward and Parkes, 1994), and cognitive capture/tunneling (Gish and Staplin, 1995; Pausie, 2015; Tufano, 1997; Ward and Parkes, 1994), and, further aggravate driver information processing. In order to avoid these negative consequences, only the information necessary for the driver should be carefully selected and displayed. In this regard, understanding the drivers' information needs, and, defining the functional requirements of automotive HUDs (what information should be displayed and when) accordingly is crucial for the design of useful HUDs. In addition, the necessary information must be presented in a manner conducive to human information processing. Hence, human factors display design principles should be used as a guide to interface design. Research efforts have been made to develop knowledge for automotive HUD interface design. Some studies, for example, investigated the impacts of major design attributes (design variables) of HUDs and their geometric elements, such as color (Choi et al., 2013; Huang et al., 2013; Moon et al., 1998), display type (analog vs. digital) (Huang et al., 2013; Moon et al., 1998), layout (Park et al., 2012) and display location (Chao et al., 2009; Flannagan et al., 1994; Morita et al., 2007; Horrey et al., 2004; Tangmanee et al., 2012; Tsimhoni et al., 2001; Tretten et al., 2011; Yoo et al., 1999). Multiple literature reviews have been conducted focusing on automotive HUD interface design (Gish and Staplin, 1995; Harrison, 1994; Tufano, 1997; Ward and Parkes, 1994).

Nielsen's notions of utility, usability and usefulness (Nielsen, 1994) may be useful in understanding the roles of functional requirements analysis and interface design mentioned above and their interplay in the design of automotive HUDs. Nielsen defined utility as the degree to which a system addresses the user's needs. Thus, the outcome of functional requirements analysis, that is, the specification of automotive HUD functions, directly affects the system utility perceived by the drivers. Usability is defined as a quality attribute that assesses how easy user interfaces are to use. The usability of the system is, therefore, mainly determined by the HUD interfaces resulting from screen-level interface design. Utility and usability together determine the overall usefulness of a system. As mentioned earlier, previous research studies have portrayed the usefulness of automotive HUDs as improving driving performance and safety, and creating new driver experience. Utility and usability are closely inter-related in the design of automotive HUDs - functional requirements analysis sets the goal and contexts for interface design, and interface design, when informed by the human factors engineering and HCI knowledge, can limit the range of realizable HUD functions. Also, both functional requirements analysis and interface design are informed and/or limited by design inputs, including drivers' characteristics, driving tasks, environments, and available vehicle technologies.

Despite the three decades of automotive HUD use and the significant previous research efforts, however, the design of useful automotive HUDs remains a challenging problem. One of the difficulties developers and researchers are experiencing is the lack of integrated understanding of the existing knowledge and views concerning both the utility (functional requirements) and usability (interface design) aspects of the automotive HUD design. While much work has been conducted, the existing knowledge and data appear disjointed and poorly integrated. Remedying this situation would greatly contribute to the design of useful automotive HUDs. It would also help identify future research directions.

As an attempt to alleviate the aforementioned problem and, therefore, contribute to the design of useful automotive HUDs, the objectives of the current study were to 1) provide an integrated understanding of the existing knowledge and views on the functional requirements (what information should be displayed and when) of automotive HUDs, and, on the basis of such understanding, 2) suggest directions for future automotive HUD design research. The aspect of interface design was not included within the scope of this review article.

According to the existing literature on the topic of functional requirements analysis, the functional requirements of a product can be largely derived from the following information sources: market research (gathering and analyzing existing product information) (Pahl and Beitz, 2013; Pugh, 1991; Sommerville and Sawyer, 1997; Sudin et al., 2010; Ulrich and Eppinger, 2011), domain expert knowledge (proposals and research results on new applications of technological products found in academic papers) (Pahl and Beitz, 2013; Pugh, 1991), and users' opinions on their needs (Sommerville and Sawyer, 1997; Sudin et al., 2010; Ulrich and Eppinger, 2011). Therefore, the current study entertained the following research questions so as to accomplish the study objectives:

- Research Question 1) What types of information are presented by the existing commercial automotive HUD systems and for what situations?
- Research Question 2) What types of information have previous studies suggested for automotive HUDs and for what situations?
- Research Question 3) What types of information do drivers require for automotive HUDs and for what situations? What is their relative importance?

Fig. 1 depicts the conceptual framework that informed the three research questions above; the conceptual framework also guided the entire research study. It shows the relationship between functional requirements analysis and interface design, and the information sources for each aspect of design. It also describes how HUD design affects the utility, usability and usefulness of the resulting system.

The first research question was addressed by examining the major automakers' automotive HUD products. The second, by reviewing academic research studies that proposed various automotive HUD functions. Finally, the third, by examining previous research studies that surveyed drivers' HUD information needs. For the second and third research questions, the systematic literature review method was employed as it is considered the best method for integrating existing knowledge on a research topic (Mulrow, 1994).

The rest of the article is organized as follows: Section 2 provides the details of the literature review processes. Section 3 presents the study results pertaining to Research Questions 1, 2 and 3. Section 4 presents discussion on the study findings. Finally, Sections 5 and 6 provide future research directions and conclusions.

2. Method

In this study, two separate literature searches were conducted: one for addressing Research Question 1, and, the other, Research Questions 2 and 3.

The literature search for Research Question 1 targeted documents describing HUDs of fifteen automobile manufacturers: Audi, Bentley, BMW Group, Ford, General Motors, Honda, Hyundai/KIA, Jaguar Land Rover, Mazda, Mercedes-Benz, PSA Peugeot Citroen, Renault, SAAB, Toyota, and Volvo. These fifteen manufacturers were the major commercial vehicle manufacturers. This study did not consider HUDs in concept cars or prototype HUDs as they did not necessarily represent the final commercial products and also it was difficult to find product descriptions for them.

For each manufacturer, the HUD systems installed in its models were identified through web searches. Then, other details, including the

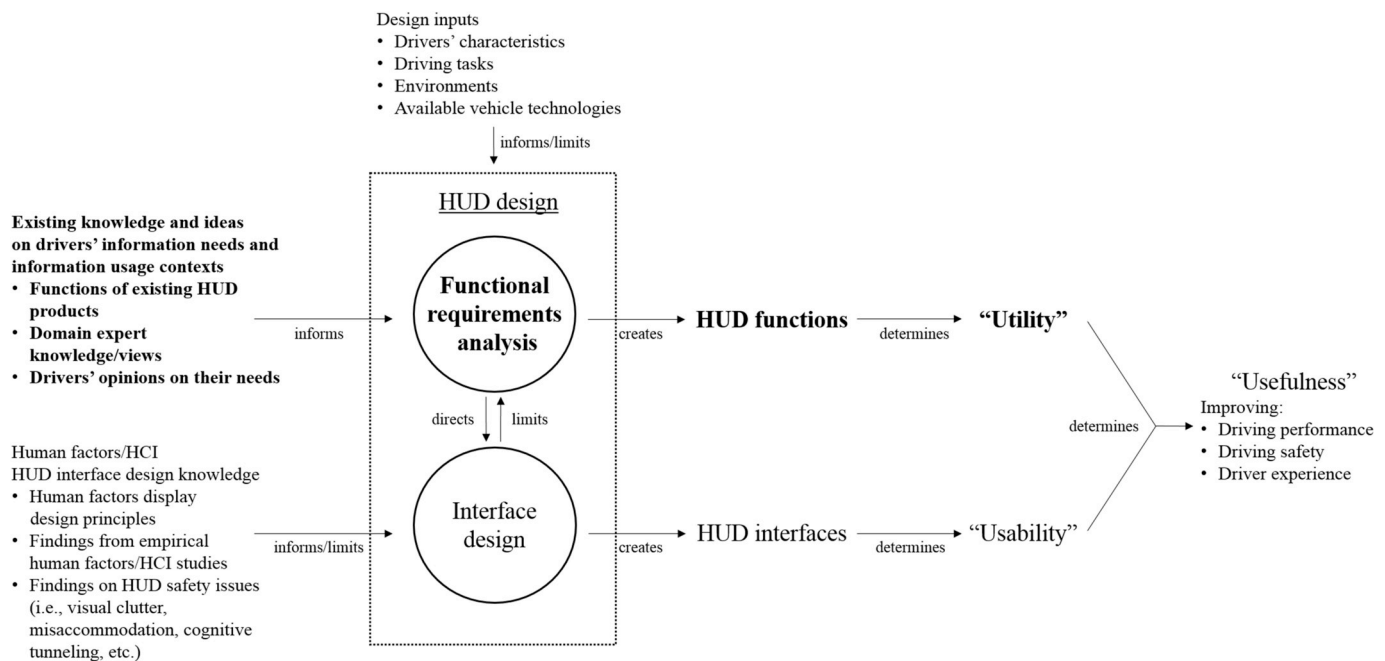


Fig. 1. Conceptual framework used to inform the research questions and guide the study (the scope of this study is represented in bold.).

specifications of the HUD systems and the contexts and purposes of information use, were examined using the vehicle manuals and available YouTube or other video clips on the internet.

The literature search for Research Questions 2 and 3 was intended to provide a comprehensive literature review. The search period was from January 1994 to September 2016. Four online databases were utilized: ACM digital Library, Science Direct, Scopus, and Web of Science. Three concepts were initially selected as the keywords for literature search: HUD, automobiles, and information display design. Then, for each initial keyword, interchangeable and topically related terms were further explored to determine more keywords. The final set of search keywords used for the literature search is shown in Table 1. The search formula used for the database searches first combined the keywords within each concept (initially chosen keyword) with the Boolean operator 'OR' and, then, linked the resulting expressions corresponding to the three concepts with the Boolean operator 'AND'. The keywords in Table 1 were generic, and, allowed identifying a wide range of documents related to the automotive HUD design.

A total of 1378 documents were obtained as a result of the keyword searches in the four databases: 138 from ACM Digital Library, 92 from Science Direct, 297 from Scopus, and 851 from Web of Science, respectively. A wide range of documents, including journal articles, conference papers, and other forms of publication, such as master's and doctoral dissertations, and technical reports were collected. For each of the 1378 documents, its title, abstract and keywords were examined with the following exclusion criteria: (1) studies that are not related to automotive HUDs, (2) duplicate studies, (3) no full-text access

Table 1
Search keywords used for literature search.

Concept	Search keywords
HUD	Head up display*, Head-up display*, HUD*
Automobiles	Automotive, vehicle*, car*, automobile
Information display design	Interface, augmented reality, information, design, human factors, system

Note: An asterisk () at the end of a keyword indicates that all terms that start with that root were included in the search.

supported, and (4) studies that are not written in English or Korean. A total of 165 relevant studies remained after the elimination of unqualified documents. Then, the 165 studies were carefully reviewed to identify the ones relevant to addressing Research Questions 2 and 3 – the studies that were irrelevant or were focused on the hardware design/development were excluded. A total of 27 studies were identified. The studies in the reference lists of the 165 studies were also examined and 17 additional studies were found. As a result, a total of 44 studies were included in this review. 41 studies pertained to Research Question 2, and, three, Research Question 3.

3. Results

3.1. Information types displayed by existing commercial automotive HUD systems (Research question 1)

A total of 27 information types were identified from examining the commercial HUDs of the fifteen automobile manufacturers. The 27 information types were grouped into five categories: vehicle state, safety, navigation, communication/infotainment and outside environment.

The vehicle state category consists of current speed, cruise control-related information, gear shift-related information, RPM, system messages, fuel-related information, high beam status, turn signal status, parking assist status, hybrid system status, race car-related information, electronic stability control status, brake assist status, eco-driving status, and tire pressure status. The safety category is composed of collision warning, road signs notification and warning, lane keeping-related information, and night vision-related warning. The navigation category consists of navigation instructions, remaining distance to destination, and compass heading. The communication/infotainment category includes radio, audio player and phone call information, and voice recognition system status. The outside environment category consists of outside temperature.

Table 2 describes the 27 information types along with their usage situations and/or purposes. As can be seen from the table, the 27 HUD information types represent those provided by traditional in-vehicle displays, such as instrument panel and navigation system displays. None of the 27 HUD information types represented novel information types created specifically for HUDs.

Table 2
Information types displayed by commercial HUDs and their usage situations and/or purposes.

Information types		Usage situations and/or purposes
Vehicle state information	Current speed	• Supporting driver vehicle longitudinal control, especially in a speed-limit zone
	Cruise control-related information	• Informing driver of cruise control system mode and setting • Preventing driver errors and supporting vehicle longitudinal control • Enhancing driver-automation interaction
	Gear shift-related information	• Informing driver of gear position and vehicle operation mode • Preventing driver errors • Suggesting the optimal gear position to improve manual driving performance
	System messages	• Informing driver of vehicle malfunctions or oncoming problems • Enabling timely vehicle maintenance • Ensuring uninterrupted driving and improving driving safety
	RPM/Tachometer	• Informing driver of vehicle's engine status • Helping drivers decide when to shift gears on a manual transmission • Improving driving performance and fuel economy
	Fuel-related information	• Indicating the need to obtain more fuel when fuel is low • Ensuring uninterrupted driving
	High beam status	• Informing driver of high beam status • Improving driver perception of outside environment • Useful when visibility is low • Improving safety
	Turn signal status	• Informing driver of turn signal status • Providing feedback to driver on his/her own action
	Parking assist system status	• Informing driver of parking assist system status • Reducing human errors • Enhancing driver-automation interaction
	Hybrid system status	• Informing driver of hybrid system status, such as power driving mode, battery status, power/torque distribution status, etc. • Applicable only to hybrid vehicles
	Race car-related information	• Informing driver of car race-related information, such as gear position and lap information • Useful for car race • Improving driving performance
	Brake assist system status	• Informing driver of brake assist system status • Reducing human errors • Enhancing driver-automation interaction
	Electronic stability control status	• Informing driver of electronic stability control status • Helping minimize the loss of control • Improving driving safety
	Eco-driving status	• Informing driver of vehicle eco-driving mode • Helping reduce fuel consumption and carbon dioxide emission
	Tire pressure status	• Enabling early detection of tire problems • Ensuring uninterrupted driving • Improving driving safety
Safety information	Collision warning (including blind spot detection, distance alert)	• Informing driver of an impending collision, and helping to prevent a collision or reduce the severity of a collision • Improving driving safety
	Road signs notification/warning	• Improving driver perception of road signs or warnings • Improving driving safety
	Lane keeping-related information	• Warning the driver of unintentional lane departures • Improving driving safety
	Night vision-related warning	• Increasing driver awareness in a dark environment • Helping detect potential hazards • Improving driving safety
Navigation information	Navigation instructions	• Providing driver with navigation instructions and information on best route • Helping drive in unfamiliar areas • Improving driving performance and safety
	Remaining distance	• Informing driver of remaining distances to the next turn, and, thereby, helping driver to know when to turn • Improving driving performance and safety
	Compass heading	• Providing compass direction to driver • Helping driver drive without navigation instructions
Communication/infotainment information	Radio-related information	• Providing driver with radio-related information, such as current radio station, station frequency, etc. • Supporting driver interaction with an in-car entertainment system • Improving driver experience
	Audio player status	• Providing driver with audio player information, such as song title, media type, etc. • Supporting driver interaction with an in-car entertainment system • Improving driver experience
	Phone call-related information	• Providing driver with phone call-related information, such as incoming calls, phone call history, etc. • Supporting driver interaction with an in-car communication system • Improving driver experience.
	Voice recognition system status	• Informing driver of voice recognition system status • Supporting driver-vehicle/driver-AI interaction • Improving driving performance and driver experience.
Outside environment information	Outside temperature	• Informing driver of outside temperature • Improving driver situation awareness (outside environment)

Table 3 summarizes the information types each manufacturer supports with its HUDs. Note that: for each manufacturer in the first row, the number in the parenthesis denotes the total number of information types that one or more of the manufacturer's HUDs display; and, for each information type in the second column, the number in the parenthesis denotes the total number of manufacturers whose HUDs (one or more) display it.

Examination of the vehicle manuals resulted in the following observations regarding the way information is presented by the commercial HUD systems:

- Each of the commercial HUD systems has a display space allocation scheme, which defines a number of sub-areas within the entire HUD display area and the information types that could be displayed in each display area. Some of the commercial HUD systems, such as Cadillac XTS 2017, Chevrolet Corvete 2018 and Acura RLX 2014, provide a few optional display layouts among which the driver can select according to the driving situation or driver need. The optional display layouts of each commercial HUD system are all based on its display space allocation scheme.
- For most of the commercial HUD systems, the display layout (or each of the optional display layouts provided) is not static but dynamic in that its content and configuration can change according to the change in situation or the occurrence of a certain event or condition; safety-related warnings and user action feedbacks could be displayed interruptively. Also, in general, for each display layout available, the driver can activate or deactivate any of the information types in it through changing the product settings. This allows for creating individual-specific display layouts.

The display layouts of some commercial HUD systems that the authors were able to find from the vehicle manuals are described in Appendix A - not all vehicle manuals provided information regarding display layouts of the HUD systems. Also, note that for the commercial HUD systems of General Motors, only two are presented since each of the others is similar to either one of the two.

3.2. Information types previously suggested for automotive HUDs by research studies (Research question 2)

Forty-one research studies identified from the literature search have proposed prototype HUD systems, each of which displayed particular types of information. They reflect the researchers' views on the functional requirements of automotive HUD. It was found that many of these information types were not supported by existing commercial automotive HUD systems.

The information types displayed by the prototype HUD systems were organized into a hierarchical structure of information categories employing the KJ method (card sorting). The resulting hierarchical structure consists of two main information categories, which are: the conventional driving-related, and the autonomous driving- and non-driving-related information category; each of the two information categories contains multiple information sub-categories, and each sub-category, lowest-level information types.

The conventional driving-related information category includes information types directly relevant to the driver tasks for maneuvering a conventional vehicle which Geiser (1985) referred to as primary driving tasks. The category consists of eight sub-categories: hazard warnings, traffic sign/signal notifications, night vision images, road visibility improvement, future state predictions, driving instructions, route planning information, and driver state/behavior feedback.

The autonomous driving- and non-driving-related information category is composed of six sub-categories: autonomous driving-related information, conventional communication-related information, driver-

to-driver communication information, driver-to-passenger/passenger-to-driver communication information, outside environment information, and entertainment contents.

Table 4 describes each of the information types examined, within the hierarchical structure of information categories and sub-categories.

3.3. Information types required by drivers (users) for automotive HUDs and their relative importance (Research question 3)

As mentioned earlier in Section 2, the literature search identified three studies relevant to Research Question 3. In each of the studies, the study participants were provided with a set of predetermined information types and were instructed to evaluate the information types in perceived importance/preference. Table 5 provides a summary of the three studies describing their major findings and study methods. The sets of information types employed in these studies differed much from one another, as shown in Table 5. The research studies did not examine the situations, tasks or purposes for which the different information types are used. Detailed descriptions of the studies are provided in what follows.

Moon and Park (1998) examined the relative importance of information employing the analytic hierarchy process (AHP) technique. Prior to the user evaluation, nine different information types were identified by analyzing conventional displays in the dashboards of existing vehicles. Information typically not presented on the dashboards, such as speed limit, pedestrian warnings, email notifications, etc., was not considered in this study. 30 male participants, ranging in age from 25 to 32 years (mean age 28.4 years), participated and all had a driver's license. The study found that fuel level was perceived to be the most important information, followed by engine overheating status, turn signal status, battery level, brake status, current speed, door open status, seat belt status and emergency light status.

Park et al. (2012) conducted a survey to determine the relative importance of various HUD information types. A total of thirty-three types of HUD information were considered and grouped into four categories: vehicle state, vehicle maintenance, navigation and surroundings awareness. The relative importance of information was determined within each information category. 34 participants aged from 20 to 49 years participated in this study. As for the vehicle state category, it was found that current speed, gearshift position and fuel level were of the highest priority, followed by RPM, HVAC system status, temperature, total distance traveled, IT devices connection status and ECO status. Regarding the information types within the vehicle maintenance category, the order of priority was as follows: brake status, tire pressure, coolant level, engine status, battery level, door open status, side mirror status, engine oil level and airbag status. Among the information types within the navigation category, speed limit was found to be the most important information, followed by remaining distance to arrival, remaining time to arrival, current location, traffic condition, turn-by-turn navigation instructions, turn signal status, and the name of destination. As for surroundings awareness, the most important information was collision warnings during parking, followed by pedestrians in the vicinity, traffic lights, locations of nearby gas stations, distances from driving/parking lanes, locations of nearby parking lots, and locations of nearby car washes.

Guo et al. (2014) surveyed 545 Chinese drivers on their attitude to automotive HUD systems. The participants (age range of 25–50 years) having a driver's license evaluated ten types of HUD information. The study found that the HUD information perceived as the most needed was gap, that is, the distance to the lead car. Current speed was ranked second, followed by traffic condition, speed of the lead car, failure notification, turn signal status, navigation, fuel level, engine speed and tire pressure. The authors recommended that the first four types of information, that is, distance to the lead car, current speed, traffic

Table 3
Information displayed by existing commercial automotive HUDs.

Manufacturers	General Motors (21)	Toyota (18)	BMW Group (15)	Honda (14)	SAAB (13)	Hyundai/KIA (10)	Mercedes-Benz (8)	PSA Peugeot Citroen (8)	Mazda (8)	Audi (8)	Volvo (7)	Jaguar Land Rover (5)	Renault (5)	Bentley (5)	Ford (2)
Brands(Models)	Buick(Lacrosse), GMC(Arcadia, Yukon), Chevrolet (Corvette, Camaro), Cadillac(XLR, ATS, XTS, CTS, XT5), Oldsmobile(Cutlass Supreme), Pontiac (Grand Prix)	Lexus (RX, HS, GS), Prius	BMW(3', 4, 5, 6, 7, X, M), MINI, Rolls-royce (Ghost)	Acura (RLX), Accord, Clarity	Saab(9-5)	Hyundai (Asian, Equus, Genesis), KIA(K9)	Mercedes-Benz(C, S, GLC)	Citroen (C6, DS 5), Peugeot (3008/5008/508)	Mazda (2, 3, 6, CX-3, CX-5, CX-9)	Audi (A6, A7, A8, S6, S7, S8, RS7)	Volvo (XC90)	Jaguar(XE, XF), Land Rover(Range Rover Sport, Evoque, Discovery Sport)	Renault Megane, Renault Samsung Motors (SM6)	Bentley (Bentayga)	Ford (Explorer, Mustang, Taurus)
Vehicle state information	Current speed (14)	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Cruise control-related information (14)	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Gear shift-related information (8)	●	●	●	●	●	●	●		●			●			
System messages (7)	●	●	●	●	●			●			●				
RPM/Tachometer (5)	●	●	●	●	●										
Fuel-related information (5)	●				●	●	●	●							
High beam status (3)	●				●			●							
Turn signal status (2)	●				●										
Parking assist status (2)		●			●										
Hybrid system status (2)		●		●											
Race car-related information (2)	●						●								
Brake assist status (2)		●							●						
Electronic stability control status (1)	●														
Eco-driving status (1)	●														
Tire pressure status (1)	●														
Safety information	Collision warning (including blind spot detection, distance alert) (11)	●	●	●	●	●		●	●		●		●		●
Road signs notification/warning (11)	●	●	●			●	●	●	●	●	●	●		●	
Lane keeping-related information (9)	●	●	●	●		●			●	●			●	●	

(continued on next page)

Table 3 (continued)

Manufacturers	General Motors (21)	Toyota (18)	BMW Group (15)	Honda (14)	SAAB (13)	Hyundai/KIA (10)	Mercedes-Benz (8)	PSA Peugeot Citroen (8)	Mazda (8)	Audi (8)	Volvo (7)	Jaguar Land Rover (5)	Renault (5)	Bentley (5)	Ford (2)
Brands(Models)	Buick(Lacrosse), GMC(Arcadia, Yukon), Chevrolet (Corvette, Camaro), Cadillac(XLR, ATS, XTS, CTS, XT5), Oldsmobile(Cutlass Supreme), Pontiac (Grand Prix)	Lexus (RX, HS, GS), Prius	BMW(3, 4, 5, 6, 7, X, M), MINI, Rolls-royce (Ghost)	Acura (RLX), Accord, Clarity	Saab(9-5)	Hyundai (Asian, Equus, Genesis), KIA(K9)	Mercedes-Benz(C, S, GLC)	Citroen (C6, DS 5), Peugeot (3008/508)	Mazda (2, 3, 6, CX-3, CX-5, CX-9)	Audi (A6, A7, A8, S6, S7, S8, RS7)	Volvo (XC90)	Jaguar(XF, XF), Land Rover(Range Rover Sport, Range Rover Evoque, Discovery Sport)	Renault Megane, Renault Samsung Motors (SM6)	Bentley (Bentayga)	Ford (Explorer, Mustang, Taurus)
Navigation information	Night vision-related warning (2)	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation	Navigation
Communication/information	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)	Instructions (14)
Outside environment information	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)	Remaining distance (7)
	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)	Compass heading (4)
	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)	Radio-related information (6)
	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)	Audio player status (5)
	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)	Phone call-related information (4)
	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)	Voice recognition system status (2)
	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)	Outside temperature (3)

*Note that: for each manufacturer in the first row, the number in the parenthesis denotes the total number of information types that one or more of the manufacturer's HUDs display; and, for each information type in the second column, the number in the parenthesis denotes the total number of manufacturers whose HUDs (one or more) display it.

Table 4
Information types suggested for automotive HUDs by past research studies and their categorization.

Main category	Sub-category & purposes	HUD information	References
Conventional driving-related information	Hazard warnings:	Lead vehicle warnings	Charissis and Papanastasiou (2008); Park and Kim (2013)
	• Supporting detecting and avoiding hazards in the environment	Lane departure warnings	Charissis and Papanastasiou (2008); Lee et al. (2015); Park and Kim (2013)
	• Improving driver situation awareness (obstacles/objects)	Pedestrian warnings	Park and Kim (2013)
	• Improving safety	Locations of possible hazards (e.g., upcoming crash)	Tonniss and Klinker (2006); Tonniss et al. (2005)
		Locations of possible hazards (e.g., other vehicles and pedestrians) and their dangerousness levels	George et al. (2012)
		Directions of possible hazards and the recommended steering wheel angle movements	Maag et al. (2015)
		Visually concealed hazards warnings	Plavšić et al. (2009)
		Blind spot image	Suzuki and Hashimoto (2012)
		Traffic sign notifications (e.g., speed limit, children, highway exits, etc.)	Lee et al. (2015); Park and Kim (2013)
		Traffic signal-related information presented when approaching an intersection (e.g., 'Prepare to stop' and 'Signals ahead')	Caird et al. (2008)
	Improving driver situation awareness (traffic system)	Traffic signal-related information in two modes (real-time upcoming traffic signal information and predicted upcoming traffic signal information considering the time to approach)	Yang et al. (2016)
	Improving safety	Highlighted infrared image of pedestrian	Tsuji et al. (2002)
		Night vision images of a pedestrian or animal	Kovordányi et al. (2006)
		*Note: This night vision system was adaptive as it was only lit up when the system detected potential hazards.	
		Night vision alerts of possible hazards and their dangerousness levels	Park et al. (2015)
		Lane marking for improving road visibility	Alexander (2005); Charissis and Papanastasiou (2008)
		Enhanced ego lane information	Biswas and Xu (2015)
		Enhanced-vision images of the road scene	Halmaoui et al. (2014); Tarel et al. (2012)
		Trajectory curve (the vehicle's future path with the current acceleration) and safety boundary curve (the vehicle's future path with full acceleration) indicator for hard cornering	Kruit et al. (2005)
		Braking distance and drive-path indicator representing the safety boundary area	Tonniss et al. (2007)
	Oncoming vehicle's future virtual projected path of 3 s	Oncoming vehicle's future virtual projected path of 3 s	Tran et al. (2013)
	*Note: This system intended to help the driver make left-turns safely across oncoming traffic without a protected left-turn signal.	Upcoming traffic congestions	Charissis and Papanastasiou (2008)
		Traffic signal-related information in two modes: real-time upcoming traffic signal information and predicted upcoming traffic signal information considering the time to approach	Yang et al. (2016)
		Navigation instructions	Charissis and Papanastasiou (2008); Park and Kim (2013)
		Navigation instructions using points of interest (POI)	Lin et al. (2011)
	*Note: POIs are the places or objects that the driver might be interested in, such as nearby famous restaurants, tourist attractions, notable roadside buildings, etc.		
	Real scene-based route guidance utilizing photos obtained from the social media		Chang et al. (2015)
	Lane change instructions/recommendations for upcoming intersections or highway exists		Lee et al. (2015); Park and Kim (2013); Yoon et al. (2014)
	Subliminal visual cues enhancing the driver's awareness of traffic signs		Riener and Jeon (2012); Riener and Thaller (2014)
	*Note: The subliminal visual cues were represented as briefly flashed visual stimuli.		Maag et al. (2015)
	Directions of possible hazards and the recommended steering wheel angle movements		

(continued on next page)

Table 4 (continued)

Main category	Sub-category & purposes	HUD information	References
Autonomous driving- and non-driving-related information	Route planning information: - Supporting vehicle route planning - Improve driver decision making - Improving driving performance	Estimated times to destination for alternative routes	Chiu and Joseph (2008)
	Driver state/behavior feedback: - Supporting detection of driver problems - Improving driver skills - Improving driving performance and safety	Driver's state alert *Note: This system provides implicit and ambient visual feedback on the driver's state (drowsiness and distractions) in the driver's viewing direction Verbal corrective advice on driving style for safe and economical driving (e.g., 'Try to avoid sudden movements of the steering wheel.' Real-time, quantitative driving behavior information for encouraging fuel-efficient and safe school bus driving (amount of engine idle time, current acceleration/deceleration rate, and miles per gallon) Real-time electric vehicle driving advice for promoting efficient driving (energy consumption scales, and coaching advice icons for stabilizing velocity, smoothing out acceleration/deceleration, and avoiding hydraulic brake usage) Automated highway information (vehicle operation mode, vehicle-to-vehicle gap, and current lane position) Partially-automated vehicle status information about the longitudinal and lateral control Notification of an imminent handover of control for Level 3 autonomous vehicles Communication information for cooperative driving among highly automated vehicles (e.g., lane change action arrows, gap information, and cooperation partner information) Phone call-related information (incoming call notification, remaining time to answer the phone under a suitable situation, and voice message-related information), and SMS and email arrival notifications Driver's sign of appreciation for another driver's behavior (a thumb-shaped icon) *Note: A gesture-based HUD system was developed. Communication information for cooperative driving among highly automated vehicles (e.g., lane change action arrows, gap information, and cooperation partner information) Gaze visualization that indicates where the passenger is looking at (for supporting the driver-passenger collaboration)	Beyer et al. (2010) Karvonen et al. (2006) Pace et al. (2007) Jagiellowicz-Kaufmann et al. (2015)
	Autonomous driving-related information: - Supporting human interaction with automated systems (automated highway, autonomous vehicles, cooperative driving) - Improving driving performance and safety		Cha and Park (2006)
	Conventional communication-related information: - Supporting driver communication with other people - Improving driver experience		Wulf et al. (2015) Politis et al. (2015)
	Driver-to-driver communication information: - Supporting driver communication with other drivers - Improving driving performance and safety - Improving driver experience		Zimmermann et al. (2014)
	Driver-to-passenger/passenger-to-driver communication information: - Supporting driver-passenger communication - Improving driver and passenger experience		Charissis et al. (2007)
	Outside environment information: - Providing driver with information about the things in the environment - Improving driver experience		Wang et al. (2014)
	Entertainment contents: - Reducing the boredom of driving and improving drive engagement - Improving driving experience (fun)	Location of an outside target that the driver points at while holding the steering wheel *Note: A gesture-based HUD system was developed. Video game-related information *Note: Two games, 'rewards of glory' and 'zombies on the road,' were illustrated as examples.	Zimmermann et al. (2014) Maurer et al. (2014) Fujimura et al. (2013)
			Schroeter et al. (2014); Steinberger et al. (2015)

Table 5
Information types required by drivers (users) for automotive HUDs and their relative importance.

References	HUD information	Research methods
Moon and Park (1998)	Fuel level Engine overheating status Turn signal status Battery level Brake status Current speed Door open status Seat belt status Emergency light status Vehicle state	● 30 participants (25–32 years) were recruited. ● Only the information types of conventional HDDs were considered.
Park et al. (2012)	Current speed, Gearshift position, Fuel level RPM HVAC system status Temperature Total distance traveled IT devices connection status ECO status Brake status Tire pressure Coolant level Engine status Battery level Door open status Side mirror status Engine oil level Airbag status Speed limit Remaining distance to arrival Remaining time to arrival Current location Traffic condition Turn-by-turn navigation instructions Turn signal status Name of destination Collision warnings during parking Pedestrians in the vicinity Traffic lights Location of nearby gas stations Distances from driving/parking lanes Locations of nearby parking lots Locations of nearby car washes Vehicle maintenance Navigation Surroundings awareness	● 34 participants (20–49 years) were recruited. ● The information types were divided into four categories, and the relative importance of information was determined within each information category.
Guo et al. (2014)	Distance to the lead car Current speed Traffic condition Speed of the lead car Failure notification Turn signal status Navigation Fuel level Engine speed Tire pressure	● 545 participants (25–50 years) were recruited. ● 7.24% of 545 participants had experience of using automotive HUDs.

*Note: The HUD information in each cell of the second column is in the order of importance, and the highest priority information is in bold.

condition, and speed of the lead car, should be considered high-priority for HUDs.

4. Discussion

4.1. What types of information are presented by the existing commercial automotive HUD systems and for what situations? (Research question 1)

As shown in Table 2, the existing commercial HUD systems collectively display a variety of information types to improve driving performance, safety and driver experience – a total of 27 information types were identified. One notable observation from Table 2 was that none of the 27 information types were created specifically for automotive HUDs – they represent information types currently provided by conventional in-vehicle displays (dashboard and navigation system displays), and, thus, have identical usage situations and purposes (Table 2).

Related to the above observation, it should be noted that in most of the current HUD-equipped vehicles, the information displayed by the HUDs is also displayed by the conventional in-vehicle displays. This redundancy or duplicate presentation seems to suggest that the manufacturers conceptualized their HUD systems as an additional complement for the conventional displays.

The conceptualization of HUDs as a complement for the conventional in-vehicle displays may be reasonable as the information types displayed by the conventional displays represent some of the key information for driving. Allocating them within the easily accessible forward FoV would likely benefit driving as long as the HUD system's interface was designed to achieve a high usability - note that a product's overall usefulness is determined by both its utility and usability (Nielsen, 1994), as depicted in Fig. 1. However, some questions arise as to the role of automotive HUDs within the continuously evolving in-vehicle displays system:

- Among the various information types of the conventional displays, which ones should be displayed redundantly by both a HUD and a conventional display and which ones, by either one of the two? In other words, what is the best way to allocate the information types of the conventional in-vehicle displays between the two display areas?
- Should HUDs be utilized only as a complement for conventional in-vehicle displays? Could they be utilized to display novel and/or non-conventional information types to enhance driving experience? Also, what are the possible roles of HUDs in future vehicles?

Table 3 shows the information types that each manufacturer supports with its HUDs. It was found that some of the 27 information types were commonly supported by majority of the automobile manufacturers - current speed, cruise control, and navigation instructions were supported by 14 out of the 15 manufacturers; also, collision warning and road signs notification/warning were supported by 11 manufacturers. The commonality seems to indicate a view shared by the manufacturers that one major function of automotive HUDs is to support primary driving tasks, such as vehicle longitudinal control, navigation and detection of safety hazards.

Aside from the commonality, Table 3 also revealed that the manufacturers varied substantially in the number of vehicle models equipped with a HUD system and in the set of information types their HUDs display. General Motors, which pioneered the adoption of the HUD technology in the automotive industry, offered a HUD to eleven of its vehicle models; and, combined together, their HUDs displayed the most (21) information types. On the other hand, Ford had three HUD-equipped vehicle models, which presented the smallest number (2) of information types. General Motors, Toyota, BMW Group, Honda and SAAB supported relatively many (13–21) information types while Jaguar Land Rover, Renault, Bentley and Ford, a small number of (2–5) information types.

It is not clear what gave rise to the observed differences between the automobile manufacturers in the set of information types supported. It would be interesting to understand how each manufacturer determined the types of information that its HUD systems support; however, gathering information on the manufacturers' in-house research activities and findings is extremely difficult, if not impossible. The observed differences may perhaps reflect the differences in the automobile manufacturers' product differentiation strategies. Another possibility is that the manufacturers have different views on the range of useful HUD functions that benefit driving, that is, the role of automotive HUDs within the in-vehicle displays system.

Related to the differences between the manufacturers described above, it is thought that the range of information types a HUD system displays would affect its utility and usability in a different manner. Increasing the range of information types would tend to enhance a HUD system's utility if the information types indeed addressed the drivers' actual information needs/wants in their driving contexts. However, a design decision to support more information types with a HUD system inevitably leads to increased system complexity, which in turn increases the difficulty of creating a user interface with high usability. It is thought that such a trade-off relationship between utility and usability needs to be taken into account with great care when determining the set of information types to be supported by an automotive HUD system. While multiple previous studies (Gish et al., 1999; Horrey and Wickens, 2004; Liu et al., 2004; Medenica et al., 2011; Nwakacha et al., 2013; Palinko et al., 2013; Steinfeld and Green, 1995; Weinberg et al., 2011) demonstrated the advantages of displaying information in the driver's FoV using HUDs, other studies (Gish and Staplin, 1995; Pauzie, 2015; Tufano, 1997; Ward and Parkes, 1994) reported potential safety concerns associated with the use of HUDs, such as visual clutter and

cognitive capture. Therefore, increasing the number of supported information types at the cost of reduced interface usability could become detrimental to driving safety. The impacts of design decision on the utility, usability and overall usefulness must be evaluated.

Good interface design based on the display design principles (Wickens et al., 2003), to some extent, would enable supporting a variety of information types within an automotive HUD system without creating the safety problems mentioned above or exceeding the human information processing capacities. However, how to accomplish that is not well understood. Currently, it is unknown what type of interface is the best for automotive HUD systems and what and how much information can be safely and effectively displayed by automotive HUDs (Gish and Staplin, 1995; Pauzie, 2015; Tufano, 1997; Ward and Parkes, 1994); no detailed interface design standards/guidelines specific to automotive HUDs are currently available. Overall, given the lack of detailed design guidance, it is thought that designing a useful HUD system requires the following efforts: 1) determining the information types that represent actual information needs of the drivers and their usage contexts, and, 2) creating different design alternatives that vary in the number of supported information types (selected among those representing actual driver information needs) and the interface design, and, comparatively evaluating them in terms of utility, usability and overall usefulness.

This study reviewed the way the commercial HUD systems presented different information by examining available vehicle manuals (see Appendix A). Each of the commercial HUD systems utilized a single or multiple display layouts based on a particular HUD display space allocation scheme. For most of the commercial HUD systems, the display layout(s) was not static but dynamic in that its content and configuration could change according to the change in situation or the occurrence of a certain event or condition; safety warnings or user action feedback could be displayed interruptively.

The idea of using a particular display space allocation scheme can be beneficial if it is designed to be compatible with the drivers' expectations or mental models. Also, providing multiple optional display layouts and event-driven, interruptive displays seems to be a solution to the problem of presenting a wide variety of information types within a HUD system.

Again, as described above, the existing commercial HUD systems differed in the range of information types supported and the interface design. Naturally, they would differ in product utility, usability and usefulness. However, such differences are currently not well understood - little research seems to have been conducted to compare existing commercial HUD systems in some measures of HUD utility, usability and/or usefulness. Related to this, it should be pointed out that measures of automotive HUD utility and usefulness currently do not seem available at least in scholarly articles while previous research studies have defined different usability measures focusing on the primary and secondary task performance during driving (Gish et al., 1999; Horrey and Wickens, 2004; Liu et al., 2004; Medenica et al., 2011; Nwakacha et al., 2013; Palinko et al., 2013; Steinfeld and Green, 1995; Weinberg et al., 2011).

The existing commercial HUD systems were also found to enable the drivers to create individual-specific display layouts by allowing them to activate or deactivate any of the information types in a display layout through changing the product settings. This capability seems highly beneficial as it allows maximizing usefulness of HUD systems at the individual driver level and for each particular driving situation, and, thereby, improves the overall usefulness of the product at the population level. One potential problem, however, might be the costs (time and efforts) involved in changing the display setting. Design solutions for minimizing the display setting costs would be needed in order for the drivers to fully utilize the feature.

4.2. What types of information have previous studies suggested for automotive HUDs and for what situations? (Question 2)

As summarized in Table 4, previous research studies have proposed displaying various information types through HUDs. Two main information categories emerged from them: the conventional driving-related information category, and the autonomous driving- and non-driving-related information category. The conventional driving-related information category included the following sub-categories: hazards warnings, traffic sign/signal notifications, night vision images, road visibility improvement, future state predictions, driving instructions, route planning information, and driver state/behavior feedback. The autonomous driving- and non-driving-related information category consisted of the following sub-categories: autonomous driving-related information, conventional communication-related information, driver-to-driver communication, driver-to-passenger/passenger-to-driver communication information, outside environment information and entertainment contents.

As for the information types in the conventional driving-related information category, a significant portion of them pertained to improving driving safety – for example, see the information types in the ‘hazards warnings’, ‘traffic sign/signal notifications’, ‘night vision images’, ‘road visibility improvement’, and ‘future state predictions’ subcategories. The prototype HUD systems displaying these safety-related information types aimed at enhancing the driver’s situation awareness (SA) at some or all of the three levels of SA: perception, comprehension, and projection (Endsley, 1995). Level 1 SA, that is, perception of important elements and events in the environment, was improved generally by enlarging the natural human perceptual volume of space and time by presenting information from sensors (Alexander, 2005; Biswas and Xu, 2015; Caird et al., 2008; Charissis and Papanastasiou, 2008; Halmaoui et al., 2014; Kovordányi et al., 2006; Lee et al., 2015; Park and Kim, 2013; Park et al., 2015; Plavšić et al., 2009; Suzuki and Hashimoto, 2012; Tarel et al., 2012; Tonnis and Klinker, 2006; Tonnis et al., 2005; Tsuji et al., 2002); such information could not be acquired by unassisted human sensory organs. Following up with the latest advances in the sensor technologies and linking them with HUDs would likely help designers ideate new automotive HUD functions, since the available technologies could serve as design inputs as shown in Fig. 1. The prototype HUD systems by George et al. (2012), Maag et al. (2015) and Park et al. (2015) enhanced Level 2 SA (defined as comprehension of the current situation) by providing interpretation of the current situation – they presented dangerousness levels of possible hazards and recommended steering wheel angle movements. HUD designs investigated by Charissis and Papanastasiou (2008), Kruit et al. (2005), Tonnis et al. (2007), Tran et al. (2013) and Yang et al. (2016) improved Level 3 SA (defined as projection of future status) by offering predictions of future states and events. Interestingly, these displays did not offer any interpretations – leaving the interpretation to the human driver may be appropriate unless the machine interpretation is extremely accurate and reliable. Also, too much interpretation of the environment for the driver might reduce the driver’s situation awareness.

The view that HUDs are a means for improving driving safety, implied by the above-mentioned safety-related information types, seems to be predicated upon the widely believed advantage of HUDs, that is, reduced EoRT and re-accommodation demands (Gish et al., 1999; Horrey and Wickens, 2004; Liu et al., 2004; Medenica et al., 2011; Nwakacha et al., 2013; Palinko et al., 2013; Steinfeld and Green, 1995; Weinberg et al., 2011). Also, it might be further justified on the basis of human factors display design principles, such as the principles of minimum information access cost and proximity compatibility (Wickens et al., 2003) – HUDs can reduce the information access cost by displaying the information in the driver’s FoV, and AR HUDs in

particular allow references to be displayed close to the referents affording more efficient information processing. However, as mentioned earlier, the use of HUDs could also create a new set of problems to the drivers, such as masking of external targets, visual clutter, misaccommodation, and cognitive tunneling, and, may adversely affect driving safety in certain situations (Gish and Staplin, 1995; Pauzie, 2015). Thus, newly proposed HUD functions (including those purporting to improve driving safety) and their interfaces must be evaluated in terms of the risks of such potential side-effects.

The information types in the autonomous driving- and non-driving-related information category have been proposed recently, especially during the past five years (2011–2016). These new information types seem to reflect the profound shift in the meaning of automobile and driving that has started to take place at the beginning of the century. Emerging autonomous vehicle and other technological innovations are expected to transform human activities inside a vehicle (Anderson et al., 2014), and, such changes will result in a new set of user information needs and wants (see Fig. 1). Automotive HUDs may become a key for addressing some of such new needs and wants although predicting what they will eventually display and for what purposes is difficult – despite the uncertainty, however, it is expected that many attempts will be made to utilize automotive HUDs for non-driving activities, such as gaming and socializing.

The information types summarized in Table 4, which represent the researchers’ point of view, illustrate how the automotive HUD technology can be combined with others to produce potentially useful applications – all of them were the results of combining HUDs with budding and blooming technologies, such as sensors, augmented reality, artificial intelligence, the internet of things, connected cars, etc. Indeed, HUDs possess one characteristic that would make them suitable for bringing the benefits of emerging technologies to the driver: it naturally connects the driver and the technological elements designed to perceive and act upon the physical environment, in the physical environment itself.

Finally, it is perhaps worth pointing out that most of the information types shown in Table 4 have not been adopted in commercial HUD systems. While the reasons are not clear, a couple of possible explanations are suggested here: first, for some of the information types presented in Table 4, they could not be adopted in commercial HUD systems because they require currently unavailable, immature or prohibitively expensive technologies. For example, presenting oncoming vehicle’s future virtual projected path (Tran et al., 2013) or video game-related information (Schroeter et al., 2014; Steinberger et al., 2015) requires the full-windshield HUD technology, which is known to be expensive and technically difficult for the implementation in passenger cars at this time. Second, information types, such as notification of an imminent handover of control (Politis et al., 2015), and communication information for cooperative driving among highly automated vehicles (Zimmermann et al., 2014), are for high-level autonomous driving. However, such high-level autonomous driving is not part of our daily life yet. Third, some of the information types, for example, video game-related information (Schroeter et al., 2014; Steinberger et al., 2015), may not be justified in terms of its costs and benefits – while such features may improve certain aspects of driver experience, they may give rise to serious side effects, such as driver distractions. It would also be extremely difficult to integrate such information into a commercial HUD system without compromising the interface usability.

4.3. What types of information do drivers require for automotive HUDs and for what situations? What is their relative importance? (Research question 3)

The literature searches identified only three studies concerning Research Question 3 (Table 5). As mentioned earlier, in each of these

studies, the study participants were provided with a set of pre-determined information types and were instructed to evaluate them in perceived importance/preference.

The three studies were found to differ substantially in their key findings, that is, the user-perceived high-priority HUD information types (Table 5). This is not surprising when considering the differences in the research methods - the studies employed very different sets of predetermined information types, and also differed substantially in the number of study participants. The differences in the sets of pre-determined information types seem to reflect each study's unique research context, such as the time of publication and the particular design problem considered.

All in all, it is thought that the three studies have limited value in helping address Research Question 3. Two major limitations of the studies are as follows:

- The studies did not examine the usage situations/contexts of the HUD information types considered (“who needs a particular information type and when or for what purposes?”). Therefore, they provide little information concerning the design of automotive HUD systems that are capable of displaying the right kind of information at the right moment in an individually-tailored manner. The information needs and wants of HUD users are time-varying and situation-dependent and different individuals have disparate information needs and wants according to their lifestyle, interests and work tasks. The design of useful HUD systems should be therefore aimed at addressing diverse and changing needs in a flexible and intelligent manner. Indeed, currently, many of the commercial HUD systems are, to some extent, reconfigurable/customizable (Selker et al., 2002) or adaptive (Dijksterhuis et al., 2012; George et al., 2012; Kovordányi et al., 2006) (see Appendix A for detailed descriptions).
- While the studies mostly did not describe the characteristics of the study participants in detail, it appears that they did not utilize specific participant inclusion or exclusion criteria related to the prior experience of using automotive HUD systems. Given that fact that automotive HUDs have not been widely available, it is likely that only a small portion of the study participants had had any prior experience of using HUDs – for example, in Guo et al. (2014), only 7.24% of 545 subjects had experience of using automotive HUDs. This may represent a serious limitation – one may not be able to accurately judge what HUD information is important and what is not without actual experience of using automotive HUDs in the real-world driving contexts. A previous literature review by Harrison (1994) also pointed out the possible effects of prior use experience on the perception of HUDs.

Further research studies are needed to address the current lack of knowledge and data concerning Research Question 3.

5. Future research directions

Some future research directions concerning the design of automotive HUD systems were derived from the current review results. They are provided below:

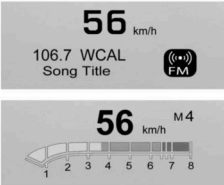
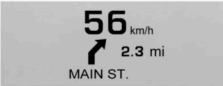
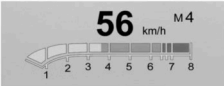
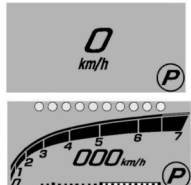
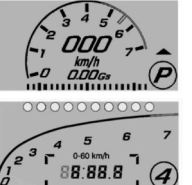
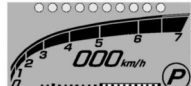
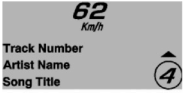
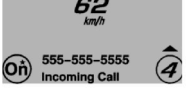
1. Future research studies should provide an established set of measures and methods for evaluating an automotive HUD system's utility, usability and overall usefulness.
 - Few research studies seem to have investigated how to evaluate the utility of an automotive HUD system.
 - While previous research studies utilized different usability measures for evaluating automotive HUD systems, no established, standard set of usability measures seems to exist at this time.
2. Research studies are needed to comparatively evaluate the existing commercial automotive HUD systems and/or the current dominant designs.
 - Both individual- and population-level measures of utility, usability and usefulness need to be defined.
 - Different automotive HUD systems exist in the market; yet, their comparative evaluation in terms of utility, usability and overall usefulness is currently unavailable. Comparative evaluation of existing design alternatives would inform design improvement and innovation.
3. Research should attempt to establish and re-establish the possible roles of HUDs in future vehicles.
 - There is a research need to define the functional requirements of automotive HUD systems for Levels 3 and 4 autonomous driving.
 - Efforts must be continually made to explore possibilities of combining newly emerging technologies (e.g., sensors, artificial intelligence and internet-of-things) with automotive HUDs.
4. Research efforts should be made to understand the actual product use practices and subjective experiences of the existing automotive HUD system users.
 - Future HUD system development must take into account the dynamic, context-sensitive and individual-specific nature of the driver information needs.
 - Little contextual inquiry and analysis research seems currently available concerning the actual automotive HUD system use practices and problems/challenges.
 - HUD information needs and design improvement points perceived by actual HUD users need to be investigated.
 - User studies on automotive HUD systems should recruit participants with a diverse range of prior HUD use experience, spanning novice to more experienced HUD users.
5. Research is needed to better accommodate different HUD users' information needs and preferences.
 - Different individuals would have disparate information needs according to their lifestyle, interests and work tasks. There is a need to examine how various user characteristics, such as lifestyle, interest, gender, age, region, etc., affect user information requirements.
 - The costs (time and efforts) involved in changing the product settings according to an individual user's unique information needs or preferences need to be minimized through good interface design or the use of machine intelligence.

6. Conclusions

In an effort to address the question of what information should be presented to the driver by automotive HUDs and when, this study investigated the developer, researcher and user perspectives on the functional requirements of automotive HUDs through literature research.

Considering the new HUD functions proposed in recent research studies, the authors think that automotive HUD systems have a potential to significantly improve the driver experience, especially through integrating with other technologies. However, in order to develop a HUD system that benefits driving, it is necessary to gain an accurate understanding of the diverse and changing user information needs and usage contexts. Also, more HCI and human factors research is needed to improve the user interface aspects.

APPENDIX A. Display layouts of some commercial HUD systems.

Manufacturers	Brands (Models)	Display layouts	
		Descriptions	Screen image examples
General Motors	Cadillac XTS 2017	The HUD system provides four display layouts from which the driver can select one: a) Speed view: current speed, navigation instructions, remaining distance, speed sign notification/warning, cruise control-related information, fuel-related information, collision warning, lane keeping-related information, audio player status, and phone call-related information b) Audio/phone view: radio-relation information, audio player status, phone call-related information, current speed, navigation instructions, remaining distance, speed sign notification/warning, cruise control-related information, fuel-related information, collision warning, and lane keeping-related information c) Navigation view: navigation instructions, remaining distance, compass heading, current speed, speed sign notification/warning, cruise control-related information, fuel-related information, collision warning, lane keeping-related information, audio player status, and phone call-related information d) Performance view: current speed, RPM/tachometer, navigation instructions, remaining distance, gear shift-related information, fuel-related information, collision warning, lane keeping-related information, audio player status, and phone call-related information	a)  b)  c)  d) 
	Chevrolet Corvete 2018	The HUD system provides four display layouts from which the driver can select one: a) Tour view: current speed, gear shift-related information, vehicle alerts, navigation instructions, remaining distance, audio player status, and phone call-related information b) Sport view: current speed, RPM/tachometer, gear shift-related information, race car-related information, vehicle alerts, navigation instructions, remaining distance, audio player status, and phone call-related information c) Track view: current speed, RPM/tachometer, gear shift-related information, race car-related information, vehicle alerts, navigation instructions, remaining distance, audio player status, and phone call-related information d) Timing view: RPM/tachometer, gear shift-related information, race car-related information, vehicle alerts, navigation instructions, remaining distance, audio player status, and phone call-related information Vehicle alerts, navigation instructions, remaining distance, audio player status, and phone call-related information are briefly displayed in any HUD view.	a)  b)  c)  d) 
			 ① When the audio player is activated  ② When the navigation system is activated  ③ When a phone call is connected  ④ When the vehicle alert system is activated
Honda	Acura RLX 2014	The HUD system provides five display layouts from which the driver can select one: a) Hybrid system status (power/torque distribution), and current speed b) Current speed, RPM/tachometer, and gear shift-related information c) Current speed, and compass heading or navigation instructions d) Current speed	a)  b)  c)  d)  e) 

When necessary, warning displays appear. Audio changes, voice recognition, and phone information are briefly displayed.



- ① When the collision warning system is activated



- ② When the lane keeping assist system is activated



- ③ When the vehicle alert system is activated



- ④ When the audio player is activated



- ⑤ When the voice recognition system is activated



- ⑥ When the phone call-related system is connected

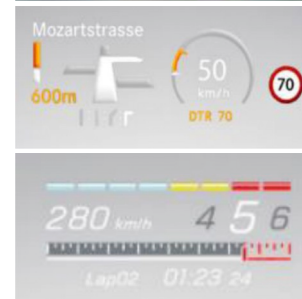
BMW Group BMW 3, 4, 5, 6, 7, X Standard display: current speed, cruise control-related information, gear shift-related information, system messages, collision warning, speed limit notification/warning, lane keeping-related information, night vision-related warning, navigation instructions, remaining distance, radio-relation information, audio player status, phone call-related information, and voice recognition system status,

BMW M BMW M display: current speed, RPM/tachometer, gear shift-related information, system messages, road signs notification/warning, navigation instructions, radio-relation information, audio player status, phone call-related information, and voice recognition system status,



Mercedes-Benz Mercedes-Benz C, S, GLC Standard display: current speed, cruise control-related information, road signs notification/warning, navigations instructions, and remaining distance

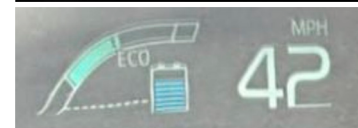
Mercedes-AMG AMG display: current speed, gear shift-related information, and race car-related information



Toyota Lexus RX 450h 2017 Lexus: current speed, cruise control-related information, gear shift-related information, RPM/tachometer, system messages, parking assist status, eco-driving status, collision warning, speed limit notification/warning, lane keeping-related information, navigation instructions, remaining distance, compass heading, radio-relation information, audio player status, and outside temperature



Prius 2017 Prius: current speed, hybrid system status, collision warning, speed limit notification/warning, lane keeping-related information, system messages, navigation instructions, and remaining distance



- Jaguar Land Rover Jaguar XE 2017
- The HUD system provides one of two display layouts according to the status of the cruise control system: a) When the cruise control system is on: current speed, cruise control-related information, speed limit notification/warning, and navigation instructions
- b) When the cruise control system is off: current speed, gear shift-related information, speed limit notification/warning, and navigation instructions



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